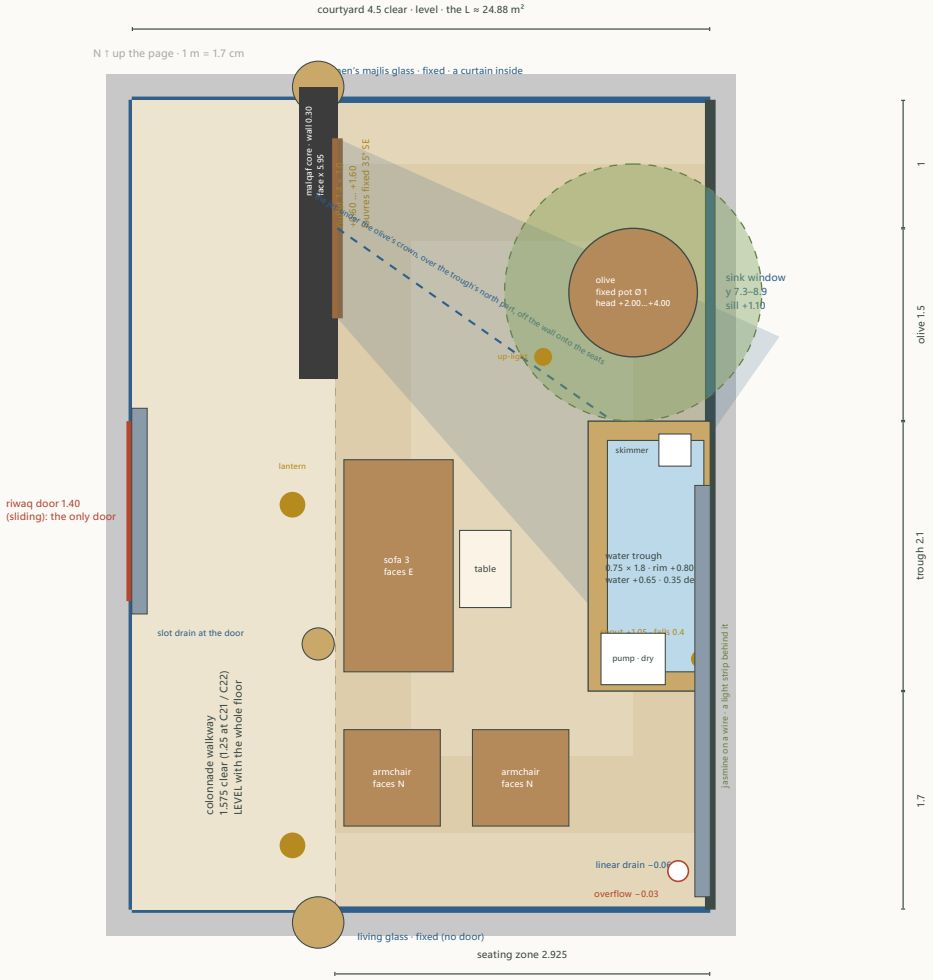
Courtyard layout · H2 ADOPTED 10 Nov 2026 · the breeze over the water: the outlet raised and turned, the trough, the olive, five seats, no TV · Taif / Palm House V10.0

The owner's choice from `studies/courtyard_haven.typ` (Phase 4) with the outlet of `studies/malqaf_outlet_B.typ` (adopted). The level floor and the drains of 3 Nov 2026 kept; the TV gone and the core's face straight (10 Nov 2026). Clear dimensions between finished faces; the data in `3d/character.yaml` `courtyard_layout` follows at the next release (the seating zone, the sink window and the drainage read from it now). Every size assumed; nothing here is for construction; the MEP engineer confirms the jet, the pump and the drains.



1 · The level floor and the paths

- **Courtyard:** 4.5 × 6.3 m between the finished faces, an L of ≈ 24.88 m² (the rectangle 28.35 less the malqaf core's 3.47 m²), **the whole floor LEVEL** (owner, 3 Nov 2026): the colonnade walkway (1.575 m to the soffit edge), the seating zone and the cleaning strips at the glass one surface with gentle falls (page 2); outdoor anti-slip porcelain (R11), the field sand, a 0.6 m honey-tone border round the seating zone, the olive's circle in the border tone; no nosing, no step.
- **Paths kept free:** the colonnade walkway (nothing west of the soffit edge x 5.925); the riwaq door (sliding 1.40, **the only door**: the living's glass stays fixed) and the living's glass (0.65 m); the women's majlis glass (1 m to the olive's pot, a **curtain inside the majlis**); 1.1 m from the armchairs to the kitchen wall; 0.6 m between the side table and the trough's rim to clean it.

2 · Five seats with backs, facing the water (H2)

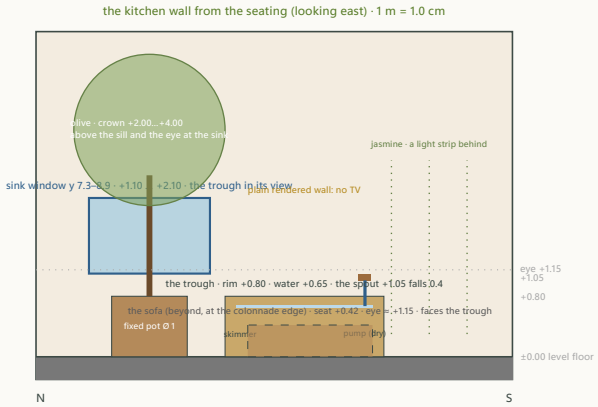
- **Modern movable pieces** (the owner chooses them), **every seat with a back, no poufs or stools**: a **3-seat sofa** 1.65 × 0.85 with its back at the colonnade edge **facing east: the water trough 1.05 m ahead, the olive and the sink window at its left**; **two armchairs** side by side on the south strip **facing north along the trough to the olive**; the **side table** 0.4 × 0.6 between the sofa and the water: **5 seats**, every seat seeing the water and the tree; the farthest seat-to-seat ≈ 2.6 m. Outdoor-rated fabrics, cushions stored in the hospitality kitchen's cupboard.
- **The olive** (owner, on apply): a 2–3 m standard in a **fixed pot Ø 1 × 0.8 m** on pot feet **in front of the kitchen sink window**, 0.1 m off the wall and centred on it (x 7.75–8.75, y 7.6–8.6): from the sink the trunk rises in the window's middle and the **crown (+2.00…+4.00, pruned flat on the wall side) stands above the sill +1.10 and the eye at the sink**, the courtyard seen either side of the trunk; ≈ 1.2 t beside the kitchen wall (a note for the engineer, close to a support); the saucer piped to the kitchen-wall drain; a drip ring on a timer; the crown's south edge 0 m from the trough: the leaves fall on the floor and at the trough's skimmer end.
- **The seating zone:** 2.925 × 5.2 = 15.21 m², less the trough 1.99 m²: **13.22 m² free**.

3 · The breeze over the water · the trough on the kitchen wall

- **The outlet** (option B, adopted): 1.4 × 1.0, raised to **+0.60…+1.60** and centred in the first arcade bay (y 6.9–8.3), the reveals splayed and the **timber louvres fixed 35° to the south-east** (owner, 11 Nov 2026, from 30°; the damper D2 behind them unchanged): at the summer-evening mean wind ≈ 1.79 m³/s leave at ≈ 2.1 m/s; the jet's axis passes 0.91 m from the olive's trunk, **under its crown** (the crown bottom +2.00 over the jet's top), and meets the kitchen wall at y 9.63: **90 % of the jet's width at the wall is over the water** (y 8.95–10.5 against the trough's 9.1–11.2); at the trough's near corner (2.6 m) the jet spans +0.15…+2.05: 66 % of its depth over the water; off the wall it spreads south along the trough and settles on the sofa and the armchairs at ≈ 0.6 m/s on the whole body (±30 %, the MEP engineer confirms), a little cooler for the water; nobody faces the jet within 3 m.
- **The trough:** 0.95 × 2.1 outside (walls 0.15) **along the kitchen wall south of the olive** (y 9.1–11.2, its north end at the crown's edge: as far from the olive as the jet's path allows), the water 0.75 × 1.8 and 0.35 deep, **rim +0.80** (seen obliquely from the sink over the sill +1.10), a **wall spout at +1.05 falling 0.4 m** at its **south end by the armchairs**: a calm, audible fall; the surface flow runs north to the **skimmer basket at the olive's end**, where the leaves fall; the pump in a dry chamber under the spout (a hatch in the rim); tanked, its drain and overflow to the rain system; the planter and the tree of 3 Nov 2026 are gone.
- **Light:** the two lanterns under the soffit edge, a warm light under the spout, the olive up-lit from the floor, a 2700 K strip behind the jasmine on the kitchen wall south of the trough; one switch by the riwaq door and one dimmer; nothing shining into the majlis or the living. **Planting:** the olive, the jasmine on a wire (y 11.3–12.6), a laurel in a smaller pot where the family likes it, out of the jet.

Study checks passed (10)

4 · The kitchen wall from the seating (the trough, the spout, the jasmine; no TV) · 5 · Drainage on a level floor · 6 · Dimensions



4 · The kitchen wall: the trough, the spout, the jasmine

- **The trough** 0.95 × 2.1 against the wall face south of the olive (y 9.1–11.2), rim +0.80: a **spout** (a short copper or bronze lip, Ø 25) at +1.05 at the **south end, by the armchairs**, gives a calm fall of 0.4 m heard from every seat (≈ 45–50 dB(A) at 2 m, assumed) and keeps the surface moving (no mosquitoes); the flow runs north to the **skimmer basket at the olive's end** (lifted and emptied by hand: the leaves and the dust); the pump (≈ 40 W, a float switch) in a dry chamber under the spout with a hatch in the rim; the trough tanked, its drain and overflow into the kitchen-wall slot drain's pipe (the rain system).
- **The sink's view:** the olive's trunk in the window's middle, its crown from +2.00 above the sill +1.10 and the eye at the sink (≈ +1.6), so the window is not blocked: the courtyard either side of the trunk, the trough's rim and water obliquely to the south (the sight line over the sill reaches +0.42 at the trough's far rim: the water seen).
- **The wall** south of the trough plain render: the **jasmine** on stainless wires (y 11.3–12.6) with a warm light strip behind it at +0.3; the armchairs 1.1 m off the wall.

5 · Drainage on a level floor

- **Falls:** the inside floors ±0.00; the courtyard finish at the three glass lines and the riwaq door at **−0.02** (a 15 mm threshold, R-08), falling **1 % away from the glass** across the floor to the kitchen wall foot at **−0.06**: no step, a 45 mm difference over the width that nobody feels.
- **Linear drains:** a slot drain with a silt basket **along the kitchen wall foot** south of the trough (y 9.6–12.8 the trough's and the pot's pipes join it) takes the whole floor; a short slot drain **at the riwaq door's threshold** (wind-driven rain at the door); the trough's drain and the olive's saucer pipe (Ø 32 in the build-up) join the wall-foot drain's pipe. **Overflow route:** a second gully at the south-east corner (rim −0.03), its own pipe east, so a blocked drain floods the courtyard 3 cm, never the rooms.
- **Pipes:** east under the kitchen floor (invert −0.55 at the drains, 1 % fall) to the **east RAIN chamber** in the yard and the soakaway (`studies/site_plan.typ`, `studies/plumbing.typ`): **never into the foul system**. **Waterproofing:** one membrane over the whole courtyard slab (lowered 0.05 for the outdoor build-up), up 0.15 behind the glass plinths and the kitchen wall base; the trough tanked separately.
- **Engineer / MEP to confirm:** the drain and pipe sizes, the rain chamber and the soakaway, the pump's circuit, the pot's point load (≈ 1.2 t), the lighting circuit and two sockets.

6 · Dimensions (clear, between finished faces)

Item	m	Note
Courtyard clear, level	4.5 × 6.3	the L ≈ 24.88 m² clear (27.19 on axes); no step
Colonnade walkway	1.575 / 1.25	to the soffit edge / at C21–C22; kept free
Seating zone (level)	2.925 × 5.2	13.22 m² free of the trough
Furniture	5 seats	a 3-seat sofa facing the water, two armchairs, a low table; movable
The olive's pot	Ø 1 × 0.8	x 7.75–8.75, y 7.6–8.6 head +2.00…+4.00 ≈ 1.2 t
Cleaning strips	0.65 S · 1 N · 1.1 E · 0.6 at the trough	living glass / majlis glass / kitchen wall / the trough's rim
Trough	0.95 × 2.1 × 0.35	kitchen wall, y 9.1–11.2 rim +0.80, water +0.65, spout +1.05 walls 0.15
Malqaf outlet (H2 / B)	1.4 × 1	y 6.9–8.3, +0.60…+1.60 louvres fixed 35° SE; 90 % of the jet's width over the water at the wall
TV	none	removed with its surround (owner, 10 Nov 2026)
Levels	−0.02 / −0.06	glass lines / wall foot; overflow −0.03 invert −0.55

What changed and where

- **3 Nov 2026:** the level floor, the enclosure on the kitchen wall. **10 Nov 2026:** no TV; the core's face straight. **H2 (adopted 10 Nov 2026):** the outlet raised and turned (`studies/malqaf_outlet_B.typ`, `studies/air_system.typ`), the trough and the olive's pot in place of the planter and the tree, five seats facing the water, the lighting and the floor pattern; **on apply (owner):** the olive in front of the sink window, the trough south of it; **11 Nov 2026:** the louvres turned to 35°; the winter sun-seat is the suite court (the first-floor plan).
- **Next release:** `3d/character.yaml courtyard_layout`, the 3D and its checks, `3d/params.yaml` (the outlet window), the DXFs, `studies/plumbing.typ`, `PROJECT_V10.md`, `DESIGN_LANGUAGE.md`.